

SURAJ DIAS

SurajDias

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ABOUT

Computer Science and Business Systems Engineering student with a strong interest in backend engineering, compiler design, and machine learning, experienced in building scalable full-stack applications.

SKILLS

Languages: Java, Python, C, JavaScript, TypeScript

Frameworks: React.js, FastAPI, Flask, Node.js

Databases: MySQL, MongoDB

Tools: Git, Linux, Postman

EXPERIENCE

BharatCares – AICTE / IBM SkillsBuild Gen AI & Cloud Computing Intern *Jun – Jul 2026*

- Independently designed and built PlacementPilot AI, a full-stack placement-preparation platform (React, FastAPI) after the originally assigned team opted out, architecting separate backend routers for resume analysis, interview preparation, and roadmap generation.
- Implemented a PDF resume-parsing pipeline (PyPDF2) with rule-based skill-gap detection and a deterministic 7-day roadmap engine, integrated Chatbase for interview practice, and deployed the app (Vercel + Render), debugging a production CORS misconfiguration.

PROJECTS

MiniCompiler Studio *React.js, Flask, Python* *Feb-Apr 2026*

- Built a compiler visualization platform with a hand-written recursive-descent parser, AST generation, and a 5-pass optimizer (constant folding, algebraic simplification, copy propagation, dead-code elimination) iterating to a fixed point over three-address code – secured Third Place in class-level PBL.
- Implemented a from-scratch regex engine that converts regular expressions into ϵ -NFAs using Thompson's construction, transforms them into minimized DFAs via subset construction and partition refinement, and visualizes each stage with animated simulations.

Career Guardian AI *React.js, FastAPI, Python* *Mar 2026*

- Built an end-to-end resume analysis pipeline that parses uploaded PDFs, extracts skills using boundary-aware regular expressions, computes an ATS-style keyword coverage score, and generates actionable recommendations for resume improvement.
- Designed an interactive career roadmap that models missing skills as a filterable dependency graph with a deterministic layout, surfacing curated learning resources for each identified skill gap.

AutoOps Pro *React.js, TypeScript, FastAPI, Python* *Apr 2026 – Present*

- Built a hybrid anomaly detection engine combining a 100-tree Isolation Forest with weighted threshold rules, fusing unsupervised and rule-based scores for explainable system health monitoring.
- Designed a weighted root-cause analysis engine normalizing CPU, memory, latency, and error-rate contributions, plus a what-if simulator evaluating six remediation strategies via real-time trend analytics.

Customer Churn Intelligence Dashboard *Python, Streamlit, scikit-learn* *Sep-Nov 2025*

- Built a customer churn prediction pipeline on telecom data, addressing class imbalance through oversampling and using a sigmoid-calibrated Random Forest to generate probability-based churn risk scores.
- Designed a feature-schema persistence system preserving one-hot encoded alignment between training and inference, with batch scoring supporting CSV and PDF report output.

EDUCATION

St. Joseph Engineering College, Mangaluru *2023 – Present*

B.E. Computer Science and Business Systems | CGPA: 7.82

CERTIFICATIONS

- Introduction to Cybersecurity** – Cisco
- Generative AI with Large Language Models** – Coursera
- Data Science Math Skills** – Coursera